

Stability and design of fixed-ended stainless steel equal-leg angle section compression members

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Abstract

The stability and design of fixed-ended stainless steel equal-leg angle section members subjected to axial compression are studied herein. An experimental investigation, comprising material testing, initial geometric imperfection measurements and tests on hot-rolled austenitic stainless steel equal-leg angle section columns is first presented. The test results, in combination with existing experimental data on stainless steel equal-leg angle section columns collected from the literature, are then used for the validation of numerical (shell finite element) models, developed within the commercial package ABAQUS. Validation is performed by means of comparisons between the test and numerical results, considering ultimate loads, failure modes and the load-deformation responses, all of which are shown to be generally in good agreement. A numerical parametric study is then presented considering angle section columns in the three main families of stainless steel (austenitic, ferritic and duplex) with a wide range of slenderness values. The behaviour and normalised load-carrying capacity of the studied members is shown to be dependent on not only the column slenderness, but also the ratio of the elastic torsional-flexural buckling load to the elastic minor-axis flexural buckling load. Finally, a design approach recently proposed for carbon steel angle section columns is extended for application to stainless steel, and verified against the experimental and numerical results. The proposed approach offers substantially improved accuracy and consistency in strength predictions compared to the existing codified design rules. The reliability of the new design provisions, with a recommended partial safety factor $\gamma_{M1} = 1.1$, is verified following the EN 1990 procedure.

Keywords: Stainless steel; equal-leg angles; fixed-ended angle columns; torsional-flexural buckling; torsional-flexural strength; EC3 design procedure

1. Introduction

Angle section compression members feature in a range of structures, including towers and trusses, and in the bracing of buildings and bridges. Although angles have been studied since the 1920s [1], their complex mechanical behaviour continues to pose challenges and current structural design provisions in international standards are known to have limitations [2, 3, 4]. New resistance functions for carbon steel angle section columns in the American [3] and European [2] design frameworks have been recently developed, leading to

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substantial improvements in the accuracy and consistency of capacity predictions.

Building on this previous work, the aim of the present paper is to develop a new approach to the design of stainless steel equal-leg angle section columns suitable for incorporation into EN 1993-1-4 [5] and to assess the accuracy and reliability of the proposals. A review of existing experimental and numerical work, as well as previous design proposals for equal-leg angle section columns, are initially described in Section 2. An experimental investigation, involving material tests and tests on five fixed-ended austenitic stainless steel equal-leg angles under axial compression, is presented in Section 3. A numerical simulation programme, performed using the commercial finite element (FE) software ABAQUS, is then presented in Section 4. The FE models are firstly validated against both the newly generated and existing physical experiments on stainless steel angle section columns, and subsequently utilised to conduct a parametric study to generate additional data, considering stainless steel angles with a wide range of slenderness and $N_{cr,TF}/N_{cr,F,v}$ values, where $N_{cr,TF}$ and $N_{cr,F,v}$ are the elastic buckling loads for torsional-flexural and minor-axis flexural buckling, respectively; the observations are summarised in Section 5. The current design rules in EC3 are evaluated using the experimental and numerical results and the shortcomings are highlighted in Section 6. New proposals for the design of fixed-ended stainless steel equal-leg angle section columns are then presented and assessed against the established data. Finally, statistical analyses are performed that confirm the accuracy and reliability of the proposed design method.

2. Review of previous research

Research into the behaviour and design of stainless steel structures has increased substantially in recent years [6], though the predominant focus to date has been on tubular [7, 8] and doubly-symmetric open sections [9, 10]. Tests on fixed-ended stainless steel angle section columns have been reported in [11, 12, 13, 14, 15, 16], as summarised in Table 1, where the manufacturing process, number of tests and range of slenderness values (with $\bar{\lambda} = \sqrt{Af_y/N_{cr,F,v}}$ and $\bar{\lambda}_{TF} = \sqrt{Af_y/N_{cr,TF}}$, where A is the cross-sectional area and f_y is the material yield strength) are presented.

Angle section columns are susceptible to a range of buckling modes. A graph showing the variation of elastic buckling load N_{cr} (for torsional (T), torsional-flexural (TF), minor-axis flexural (F_v) and major-axis flexural (F_u) buckling) with length L (on a logarithmic scale) for a typical ($100 \times 100 \times 5$) equal-leg angle section column is shown in Fig. 1. Torsional and major-axis flexural elastic buckling are always combined in angles, but for short-length angle section columns the torsional-flexural elastic buckling load approaches the torsional elastic buckling load and the buckling mode can be said to be torsional-dominant, as can be seen in Fig. 1. At intermediate lengths the critical elastic buckling mode transitions from torsional-flexural to minor-axis flexural buckling. The expressions for calculating the minor-axis flexural $N_{cr,F,v}$, major-axis flexural $N_{cr,F,u}$ and torsional $N_{cr,T}$ elastic buckling loads are as follows:

$$N_{cr,F,v} = \frac{\pi^2 EI_v}{L_{cr,v}^2}, \quad (1)$$

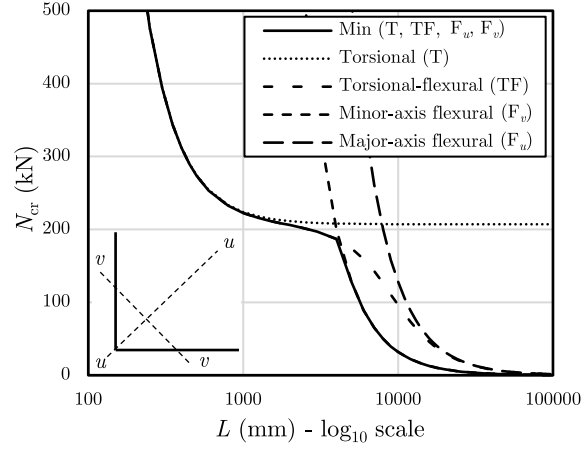


Figure 1: Elastic buckling loads N_{cr} for the different modes plotted against length L for an equal leg angle section ($100 \times 100 \times 5$) column with fixed-ended support conditions.

$$N_{cr,F,u} = \frac{\pi^2 EI_u}{L_{cr,u}^2}, \quad (2)$$

$$N_{cr,T} = \frac{1}{i_0^2} \left[GJ + \frac{\pi^2 EI_w}{L_{cr,T}^2} \right], \quad (3)$$

where E is the Young's modulus, I_v and I_u are the second moments of area about the minor and major axes respectively, $L_{cr,v}$ and $L_{cr,u}$ are the buckling lengths for minor- and major-axis flexural buckling respectively, i_0 is the cross-section polar radius of gyration with respect to its shear centre, G is the shear modulus, J is the St. Venant torsion constant, I_w is the warping constant associated with secondary warping, and $L_{cr,T}$ is the buckling length for torsional buckling (about the longitudinal axis). For isotropic materials $G = E/[2(1 + \nu)]$, with ν being the Poisson's ratio, and the definitions of i_0 , J and I_w for equal-leg angles are given in Eq. (4), where b and t are the leg length and thickness of the angle, respectively:

$$i_0^2 = \frac{b^2}{3}, \quad J = \frac{2bt^3}{3}, \quad I_w = \frac{b^3 t^3}{18}. \quad (4)$$

The expression for determining the torsional-flexural elastic buckling load $N_{cr,TF}$ is as follows:

$$N_{cr,TF} = \frac{N_{cr,F,u} + N_{cr,F,v} - \sqrt{(N_{cr,F,u} - N_{cr,F,v})^2 + 4N_{cr,F,u}N_{cr,F,v} \frac{u_0^2}{i_0^2}}}{2(1 - \frac{u_0^2}{i_0^2})} \quad (5)$$

where u_0 is the distance from the shear centre to the cross-section centroid.

Design rules for stainless steel angle section columns are included in a number of international standards, including EN 1993-1-4 [5], SEI/ASCE-8 [17] and AS/NZS4673 [18]. Liang et al. [12] and Zhang et al. [14] investigated the accuracy of EN 1993-1-4 [5], SEI/ASCE-8 [17] and AS/NZS4673 [18] in determining the ultimate capacity for hot-rolled and cold-formed stainless steel equal-leg angle section columns; highly conservative and scattered predictions were revealed. Zhang et al. [14] also studied the accuracy of the direct strength method (DSM) based design rules developed by Dinis and Camotim [3]; the strength predictions were found to be

accurate and consistent, though, owing to the fact that the provisions were established for carbon steel rather than stainless steel angles, a large portion of the results were on the unsafe side. Menezes et al. [11], Sun et al. [13], Sarquis et al. [15] and Sirqueira et al. [16] evaluated the accuracy of EN 1993-1-4 [5] in predicting the strength of stainless steel angle section columns, revealing undue conservatism. Menezes et al. [11] and Sarquis et al. [15] developed new design proposals based on EN 1993-1-4 [5]. Sun et al. [13], alongside Sirqueira et al. [16], showed that the continuous strength method (CSM) capacity predictions were more accurate and consistent than EN 1993-1-4 [5], although with some results on the unsafe side.

Table 1: Summary of existing experiments on fixed-ended stainless steel equal-leg angle section columns.

Reference	Stainless steel family	Manufacturing process	Range of slenderness $\bar{\lambda}$ or $\bar{\lambda}_{TF}$	No. of tests
Menezes et al. [11]	Austenitic	Hot-rolling	0.50 – 0.75	13
Liang et al. [12]			0.40 – 1.00	16
Sun et al. [13]			0.40 – 0.70	10
Sarquis et al. [15]			0.60 – 0.80	10
Sirqueira et al. [16]			0.75 – 1.00	18
Zhang et al. [14]		Cold-forming	0.50 – 1.25	16
Dobrić et al. [19]			1.35 – 1.40	3
Total				86

In the present study, tests are performed on hot-rolled stainless steel angle section columns that broaden the range of member slenderness $\bar{\lambda}$ values (0.40–1.40) explored experimentally, numerical models are validated and used to perform a parametric study and new structural design rules are developed. The new design proposals are evaluated against the experimental and numerical results and shown to offer substantially more accurate and consistent resistance predictions than the existing rules given in Eurocode 3.

3. Experimental investigation

3.1. General

The majority of existing experiments on fixed-ended stainless steel angles in the literature involve short-to-intermediate-length members failing by torsional-flexural buckling. The experimental programme performed herein focusses on stainless steel angles under axial compression failing by minor-axis flexural buckling. The experimental programme involved material tensile coupon tests, initial geometric imperfection measurements and tests on five stainless steel angle section columns with varying lengths and fixed boundary conditions. The tested cross-section was a hot-rolled $70 \times 70 \times 10$ equal-leg angle in Grade 1.4307 austenitic stainless steel. The chemical composition of the test material, as reported in the manufacturer’s mill certificate, is presented in Table 2.

Table 2: Chemical composition of tested cross-section from mill certificate.

Cross-section	C (%)	P (%)	S (%)	Si (%)	Mn (%)	Cr (%)	Ni (%)	Mo (%)
70×70×10	0.022	0.034	0.001	0.328	1.552	18.17	8.06	0.274

78 3.2. Tensile coupon tests

79 Prior to the column tests, material testing was conducted to obtain the material properties of the studied
 80 austenitic stainless steel angle section. Two coupons were cut from the middle portion of both legs in the
 longitudinal direction (see Fig. 2), with the coupon dimensions chosen to meet the requirements specified in

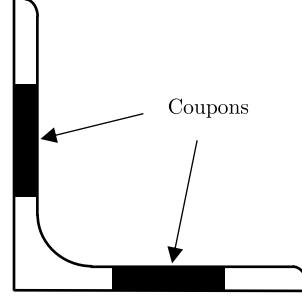


Figure 2: Location of tensile coupons within the angle section.

81
 82 EN ISO 6892 [20]. The tensile tests were conducted using an Instron 250 kN testing machine. Two linear
 83 electrical resistance strain gauges were attached to the centre of the front and back faces of the coupons at
 84 mid-height to determine the strains in the longitudinal direction. Two white dots were marked on one side
 85 of the coupons at a gauge length of 100 mm within the parallel length, and the relative movement of the two
 86 dots was tracked by a video extensometer and used to obtain the average strain over the gauge length. The
 tensile test set-up is illustrated in Fig. 3. The engineering stress–strain curves from the tensile coupon tests

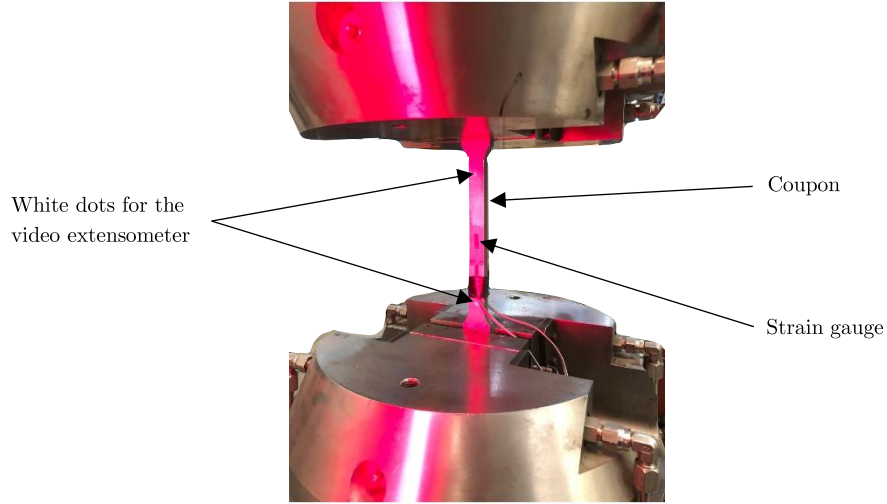


Figure 3: Tensile coupon test set-up.

87
 88 are presented in Fig. 4. The first stage of the stress–strain curves up to the yield (0.2% proof) stress f_y was
 89 obtained using the strain gauge readings, while the second stage of the curve until fracture was derived based on
 90 the video extensometer readings. The average obtained mechanical properties are presented in Table 3, where
 91 E is the Young’s modulus, f_y and $f_{1.0}$ are the 0.2% and 1.0% proof stresses, respectively, ϵ_u is the strain at the
 92 ultimate stress f_u , ϵ_f is the fracture strain measured over the 100 mm gauge length, and n , m_u and $m_{1.0}$ are
 93 the Ramberg–Osgood strain hardening exponents, with m_u and $m_{1.0}$ corresponding to the models proposed by
 94 Mirambell–Real [21] and Gardner–Ashraf [22], respectively.

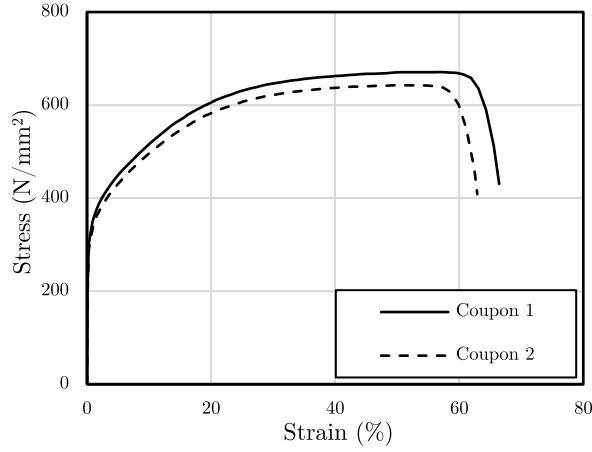


Figure 4: Measured engineering stress-strain curves from tensile coupon tests.

Table 3: Average measured material properties of the tested cross-section from tensile coupon tests.

Cross-section	E (N/mm ²)	f_y (N/mm ²)	$f_{1.0}$ (N/mm ²)	f_u (N/mm ²)	ϵ_u (%)	ϵ_f (%)	n	m_u	$m_{1.0}$
$70 \times 70 \times 10$	190000	303	356	657	53.8	61.2	11.8	3.5	2.4

3.3. Initial geometric imperfection measurements

Initial geometric imperfections in the angle section column specimens, including the bow imperfection about both the minor and major axes, denoted as δ_v and δ_u respectively, alongside the twist ϑ along the member length, were measured prior to testing. The three modes of imperfection, presented in Fig. 5, were determined

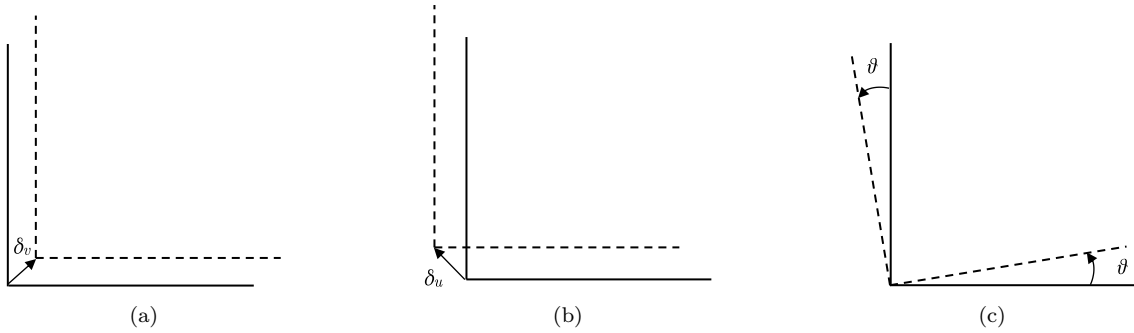


Figure 5: Illustrations of amplitudes of imperfection shapes for (a) minor-axis flexural, (b) major-axis flexural and (c) torsional modes.

using the same imperfection rig as employed by [23]. The set-up for measuring the initial imperfections is presented in Fig. 6. Four linear variable displacement transducers (LVDTs) were placed at 10 mm from the corner and the edges of the angle sections and the LVDT readings were recorded along four longitudinal lines. The four LVDTs ran on rails with an attached string pot allowing determination of the position of the LVDTs along the member length. The initial geometric imperfections were obtained using the LVDT readings as follows. The minor axis imperfection δ_v was determined from:

$$\delta_v = \frac{\delta_x + \delta_y}{\sqrt{2}} \quad (6)$$

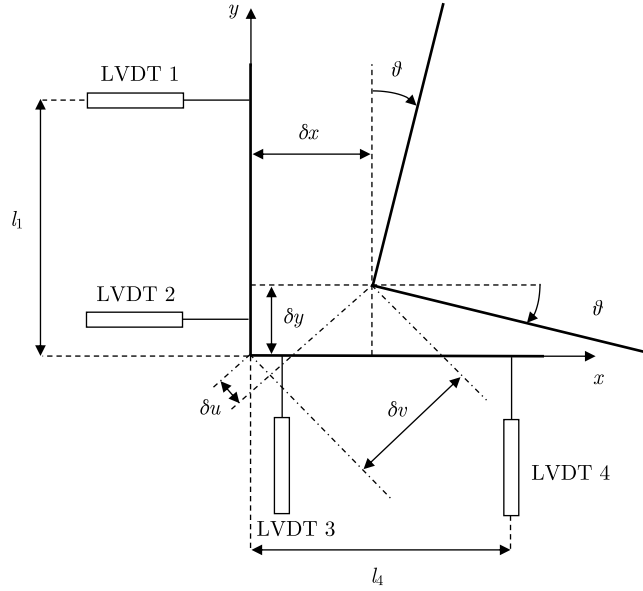


Figure 6: Location of LVDTs for initial imperfection measurement.

while the major axis imperfection δ_u was determined from:

$$\delta_u = \frac{\delta_x - \delta_y}{\sqrt{2}} \quad (7)$$

where δ_x and δ_y are obtained thus:

$$\delta_x = \frac{\delta_1 - l_4 \tan \vartheta + y_1 \tan \vartheta + x_4 \tan^2 \vartheta}{1 + \tan^2 \vartheta} \quad (8)$$

$$\delta_y = \frac{\delta_4 + l_1 \tan \vartheta - x_4 \tan \vartheta + y_1 \tan^2 \vartheta}{1 + \tan^2 \vartheta} \quad (9)$$

in which δ_1 , δ_2 , δ_3 and δ_4 are the readings of LVDT1, LVDT2, LVDT3 and LVDT4, respectively. The twist imperfection ϑ was determined from:

$$\vartheta = \left[\tan^{-1} \left(\frac{\delta_1 - \delta_2}{x_1 - x_2} \right) + \tan^{-1} \left(\frac{\delta_3 - \delta_4}{y_3 - y_4} \right) \right] / 2 \quad (10)$$

Typical measured longitudinal distributions of initial global and torsional geometric imperfections are presented in Fig. 7. A half-sine wave was employed to represent the shape of the imperfections, with the amplitude being taken as the maximum measured deviation from flatness. A summary of the measured initial geometric imperfection amplitudes of the specimens relative to the column length L is provided in Table 4. Note that a positive minor-axis bow imperfection value indicates that the imperfection direction is towards the free edges of the cross-section, as shown in Fig. 5a.

3.4. Column buckling tests

Tests on five fixed-ended hot-rolled Grade 1.4307 austenitic stainless steel equal-leg angle section columns were performed to study their buckling response and load-carrying capacity. The member lengths L were varied

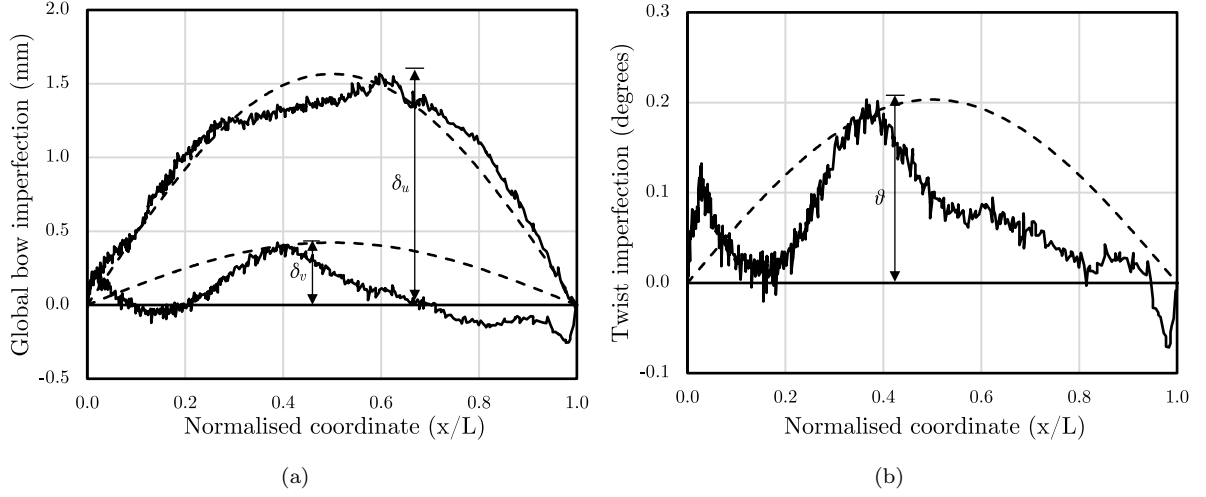


Figure 7: Typical measured initial (a) global bow and (b) twist geometric imperfections versus normalised position along the member length, with dashed lines marking representative half-sine waves.

Table 4: Summary of initial geometric imperfection amplitude measurements.

ID	L/δ_v	L/δ_u	ϑ/L (deg/m)
1	-1427	6660	0.12
2	-1762	1871	0.07
3	-2220	2498	0.06
4	-2231	3967	0.07
5	-1946	1998	0.06
Mean	-1917	3398	0.08
CoV	0.18	0.59	0.34

to cover a range of member slenderness $\bar{\lambda}$ values. The column tests were performed in an Instron 2000 kN hydraulic testing machine under displacement control with a constant loading rate of 0.015 mm/min. End plates were welded to both ends of each specimen and clamped to the base and top of the testing machine. This test set-up prevented rotation about all axes and created fixed-ended boundary conditions. Two LVDTs were used to measure the end-shortening and minor-axis lateral deflection of the columns at mid-height. Four string pots, attached to the corner and the tips of both legs of the angle section columns, were employed to measure the twist at mid-height. Six electrical resistance strain gauges and LVDT were affixed to the column specimens at mid-height to measure the longitudinal strains at the corner and the tips. The configuration of the string pots and strain gauges is shown in Fig. 8 and the column buckling test set-up is presented in Fig. 9. The key

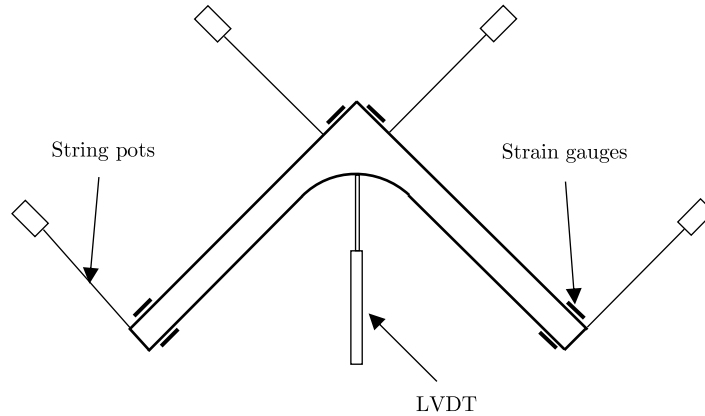


Figure 8: Configuration of strain gauges and string pots and LVDT at mid-height for column specimens.

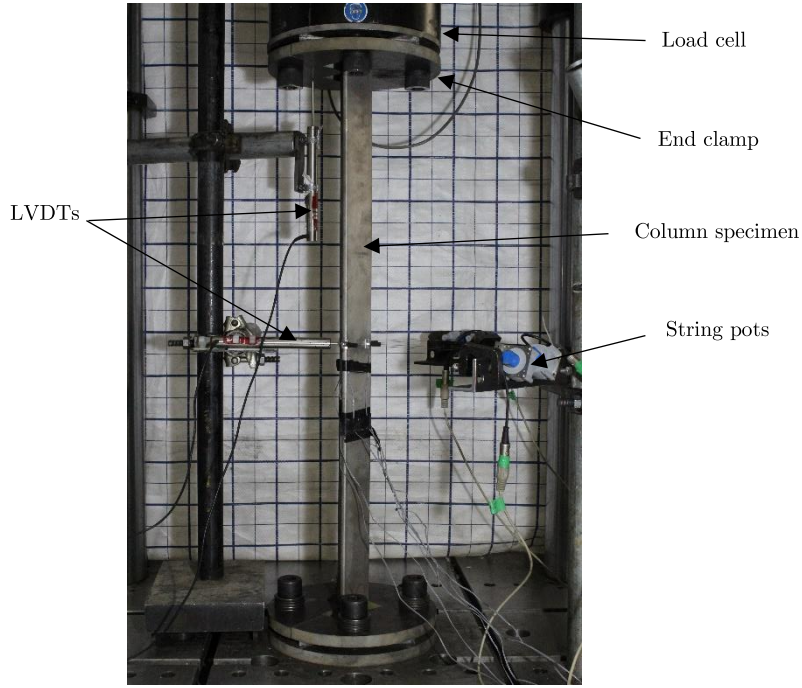


Figure 9: Column buckling test set-up.

test results, including the load, machine displacement and readings from the strain gauges, LVDTs and string pots were recorded at 0.5 s intervals using the data acquisition equipment DATASCAN and logged using the DSLOG software.

The geometry and notation for the tested equal-leg angles is presented in Fig. 10, while a summary of the

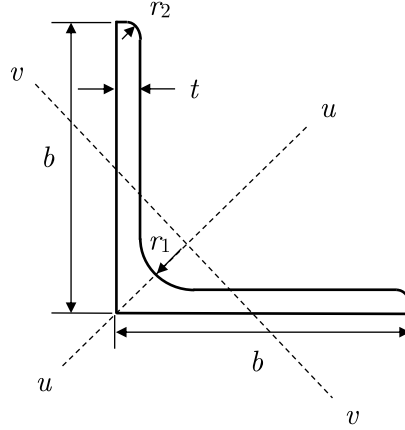


Figure 10: Geometry and notation for equal-leg angle sections.

measured dimensions and key test results is presented in Table 5. The axial load versus (a) end-shortening and

Table 5: Measured geometric properties of test specimens and key results from column buckling tests.

Cross-section	ID	b (mm)	t (mm)	r_1 (mm)	r_2 (mm)	L (mm)	N_u (kN)
$70 \times 70 \times 10$	1	70.45	9.65	7.0	3.5	999	361.2
	2	70.41	9.61	7.0	3.5	1497	314.9
	3	70.42	9.63	7.0	3.5	1998	234.2
	4	70.45	9.63	7.0	3.5	2499	207.8
	5	70.59	9.63	7.0	3.5	2997	143.8

(b) lateral deflection about the minor axis at mid-height are plotted for each test in Fig. 11. All tested columns

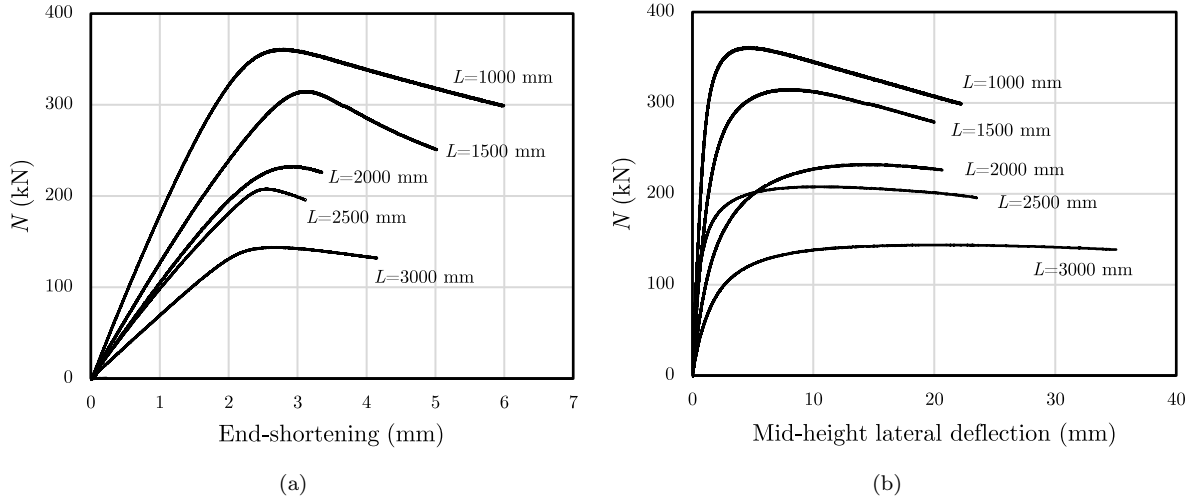


Figure 11: Axial load N versus (a) end-shortening and (b) mid-height minor-axis lateral deflection for the five test specimens, the nominal lengths of which are indicated on the figure.

failed by minor-axis flexural buckling, since $N_{cr,F,v} < N_{cr,TF}$, and the direction of buckling was dictated by the direction of the minor-axis initial global imperfections. The twist and major-axis lateral-deflection at mid-height were very small throughout all tests (less than 1 degree and less than 1 mm, respectively) and are thus not presented. A comparison of the column buckling test data with the flexural buckling curve (with limiting slenderness $\bar{\lambda}_0 = 0.2$ and imperfection factor $\alpha = 0.76$) for angle section members set out in prEN 1993-1-4 [24] is presented in Fig. 12, where $\chi = N_u / A f_y$ is the buckling reduction factor with N_u being the ultimate load

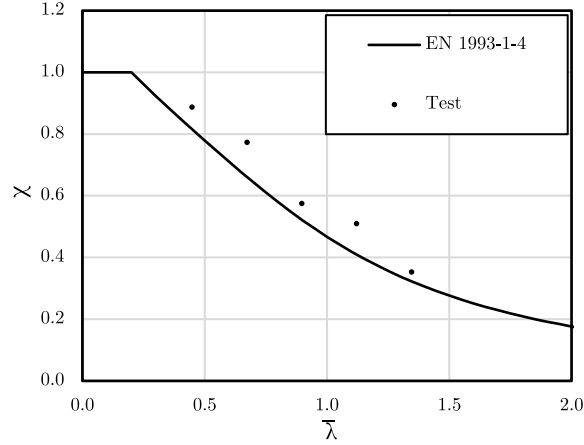


Figure 12: Comparison of column buckling test data with flexural buckling curve for angle section members set out in prEN 1993-1-4 [24].

reached during the test. The test results generally follow the trend of the buckling curve and lie consistently on the safe side.

4. Numerical modelling

The commercial software ABAQUS was used herein to create finite element (FE) models to simulate the mechanical behaviour of stainless steel equal-leg angle section members under axial compression. Validation of the FE models against the newly generated and existing experimental data on stainless steel equal-leg angle

section columns is first presented. Angle section columns with cylindrically-pinned (CP) boundary conditions, in addition to fixed boundary conditions were considered in the model validation to extend the range of considered member slenderness and to demonstrate the wider applicability of the developed models. The model validation is presented in Section 4.2. In Section 4.3, a parametric study is conducted to investigate the behaviour of equal-leg angle section columns of different stainless steel grades, produced by different manufacturing processes – hot-rolling and cold-forming – and considering a range of geometries.

4.1. General modelling assumptions

The general modelling assumptions are described in the current section. A 4-noded shell element with reduced integration, 6 degrees of freedom (3 translational and 3 rotational) at each node and hourglass control, designated S4R in the ABAQUS element library [25], was employed to model the angle section members. The chosen element is suitable for simulating the behaviour of both thin- and thick-walled cross-sections, and has been widely used for the simulation of stainless steel open-section members [26, 27, 28]. A mesh size of approximately 5 mm (typically resulting in at least 10 elements across each leg of the angles) was utilised.

Both hot-rolled and cold-formed angles were modelled. For the hot-rolled angles, the root fillet was modelled by introducing thicker elements at the junction between the angle section legs, while the thickness was offset to the outer surface of the cross-section to avoid overlapping of the two legs, which were connected using multi-point constraints (MPC), type BEAM, as shown in Fig. 13a. The widths of the elements 1, 2, 3, 4 and 5 used in forming the fillet radius in the hot-rolled sections, labelled 1, 2, 3, 4 and 5 in Fig. 13a, were $0.7t$, $0.3t$, t , $0.3t$ and $0.7t$, respectively, while the corresponding thicknesses were $1.15t$, $1.3t$, t , t and $1.15t$, respectively. The rounded corner in the cold-formed angles was represented by employing a finer mesh in the curved corner region than in the flat regions, as shown in Fig. 13b. Four elements, labelled 1, 2, 3 and 4 in Fig. 13b, with the same widths and thicknesses were used to form the curved region. To model the axial load applied to the members, a longitudinal displacement was imposed at one end through a reference point that was free to move longitudinally. The geometrically and materially nonlinear analyses were executed using the Riks arc-length method [29].

For validation of the FE models, a series of experiments on stainless steel angle section columns were simulated (see Section 4.2). The measured geometric properties and stress–strain curves were employed in the analyses. For the cases in which the full stress–strain curves were not reported, and throughout the parametric study, the two-stage Ramberg–Osgood material model [30], with measured or defined values for the key mechanical properties, was utilised. A Poisson’s ratio of 0.3 in the elastic range was assumed in all cases. The material properties were input into ABAQUS in the form of true stress σ_{true} and true plastic strain ϵ_{true} , converted from the measured engineering stress σ and strain ϵ using the following well-known relationships:

$$\sigma_{\text{true}} = \sigma (1 + \epsilon), \quad (11)$$

$$\epsilon_{\text{true}} = \ln (1 + \epsilon) - \frac{\sigma_{\text{true}}}{E}. \quad (12)$$

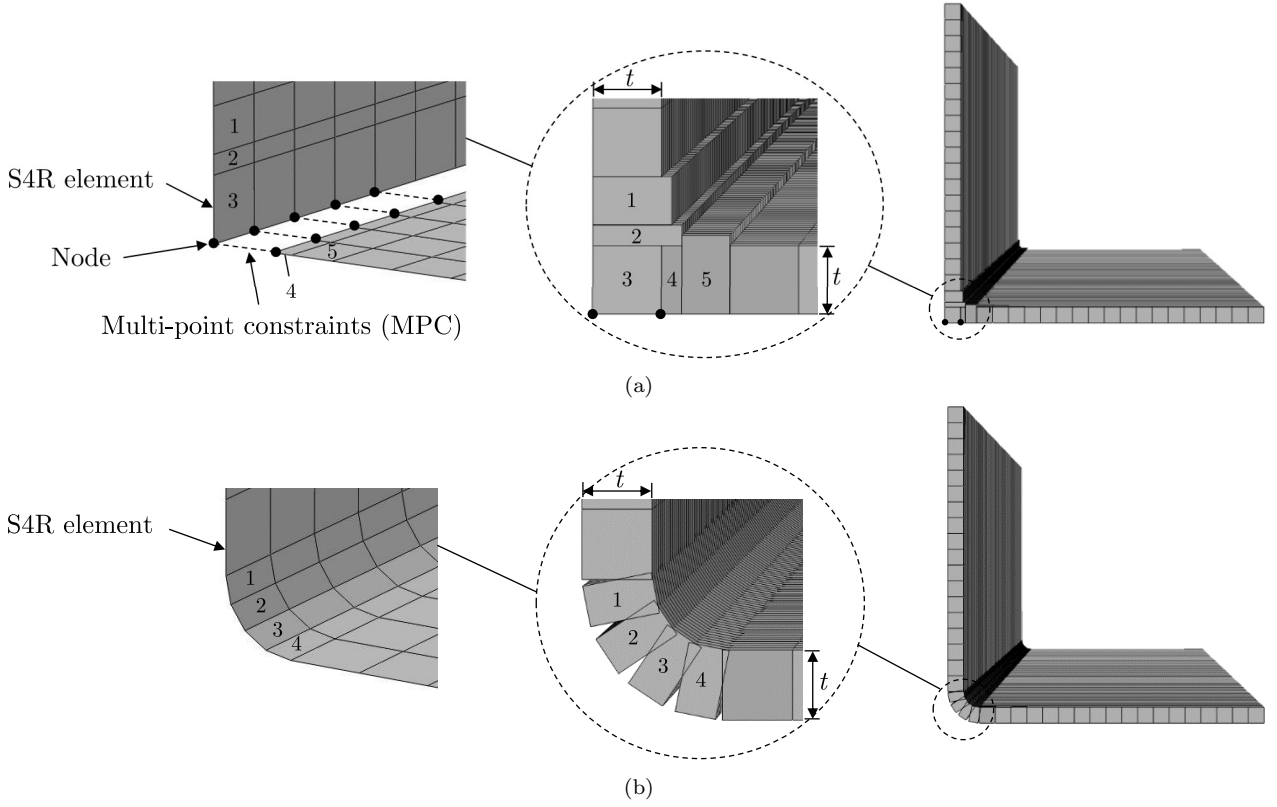


Figure 13: Representation of corner region in: (a) hot-rolled and (b) cold-formed angle section members in the FE models.

For the hot-rolled stainless steel angles, a bilinear residual stress distribution with a peak value of 70 MPa, constant with thickness and leg width, was employed [2], as illustrated in Fig. 14, where the negative sign

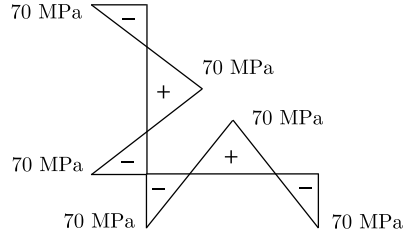


Figure 14: Residual stress distribution employed for hot-rolled stainless steel equal-leg angles in the present FE models.

indicates compressive stress and the positive sign denotes tensile stress. Experiments on hot-rolled stainless steel angle [31] and channel [32] sections have shown maximum compressive residual stress amplitudes similar to that assumed. The residual stresses were input into ABAQUS using the *INITIAL CONDITIONS command. For the cold-formed stainless steel angles, the dominant bending residual stresses were assumed to be inherently captured in the stress-strain curves [33] and were therefore not explicitly defined. For the validation study, if reported, the measured amplitude and direction of initial imperfections (including those reported in Table 4 for the columns tested in the present study) were used. However, where measured data were not available, and throughout the parametric study, a sinusoidal half wave function over the member length with an amplitude of $L/1000$ at midspan was adopted about both principal axes for the initial bow imperfections. The direction of the minor-axis bow imperfection was chosen to give the lowest resistance, which was consistently towards the tips of the cross-section legs (i.e. in the direction indicated in Fig. 15b). A similar shape was also adopted

for the initial twist with an amplitude of $\vartheta = \tan^{-1}(L/1000b)$ at midspan. The shape and amplitude of the initial bow and twist imperfections are presented in Fig. 15. This approach of using measured imperfection amplitudes in conjunction with assumed sinusoidal (or eigenmode-affine) shapes is common in the numerical simulation of structural steel [34] and stainless steel [9, 11, 13, 15] members. In all cases, the end cross-sections of the columns were flat and parallel, as shown in Fig. 16. Boundary conditions were applied to the end supports through kinematic coupling constraints to link all degrees of freedom (translational and rotational) of the member end-nodes to two reference points. Fixed-ended boundary conditions were created by restraining the necessary degrees of freedom at the reference points. By employing kinematic coupling constraints, warping was also prevented at both ends.

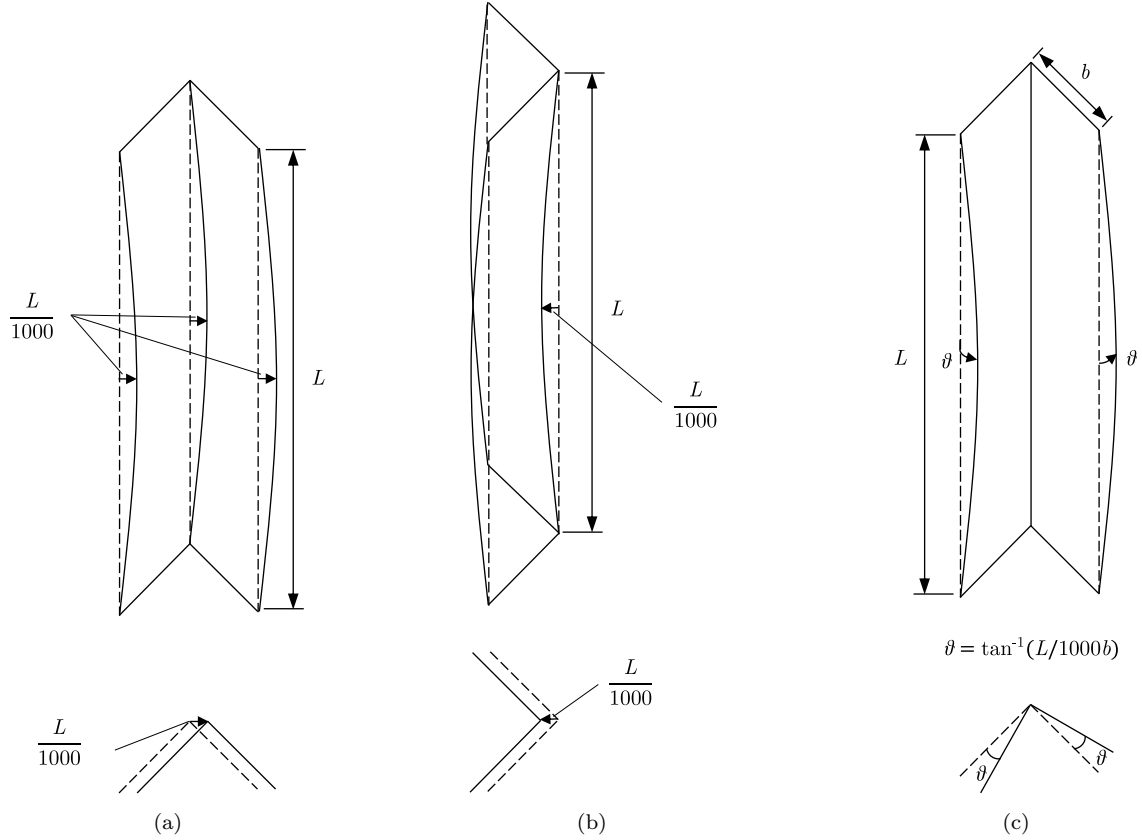


Figure 15: Initial geometrical (a) major-axis, (b) minor-axis and (c) twist imperfection shapes and amplitudes at the midspan cross-section.

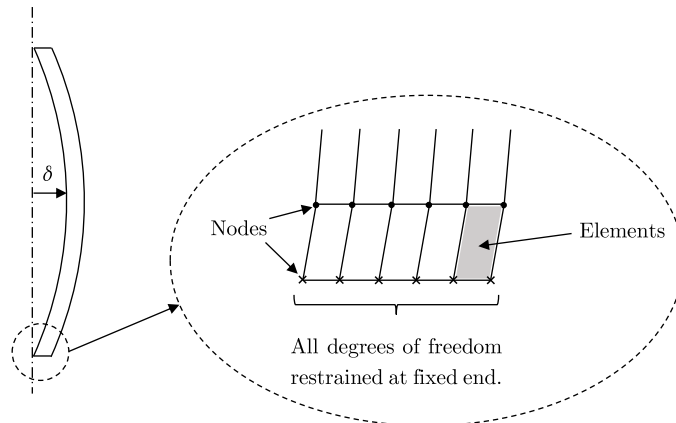


Figure 16: Modelling of end conditions in FE models of imperfect columns.

4.2. Validation

The FE models were validated against the five tests from the present study, together with a total of 122 existing experiments on stainless steel angle section columns from a wide range of sources in the literature [11, 12, 13, 14, 15, 16, 19, 35, 36, 37]; a summary of the comparisons between the FE model ultimate loads and those obtained experimentally is presented in Table 6 and Fig. 17. Overall, there is good agreement between

Table 6: Summary of comparisons of FE model ultimate loads $N_{u,FE}$ with those obtained experimentally $N_{u,Test}$.

Source	Manufacturing process	B/C	No. of tests	$N_{u,FE}/N_{u,Test}$			
				Mean	CoV	Min	Max
Menezes et al. [11]	Hot-rolling	F	13	1.04	0.06	0.96	1.16
Liang et al. [12]		F	16	1.01	0.04	0.91	1.09
Sun et al. [13]		F	10	1.00	0.07	0.92	1.12
Sarquis et al. [15]		F	10	0.98	0.07	0.90	1.09
Sirqueira et al. [16]		F	18	0.96	0.05	0.87	1.07
Zhang et al. [35]		CP	12	0.98	0.06	0.91	1.09
Zhang et al. [37]		CP	12	0.98	0.04	0.93	1.03
Present study		F	5	0.99	0.08	0.91	1.10
Zhang et al. [14]	Cold-forming	F	16	1.00	0.07	0.92	1.12
Dobrić et al. [19]		F	3	1.12	0.02	1.10	1.15
Dobrić et al. [19]		CP	8	1.02	0.22	0.71	1.29
Zhang et al. [36]		CP	4	0.99	0.05	0.95	1.05
Total			127	1.00	0.08	0.71	1.29

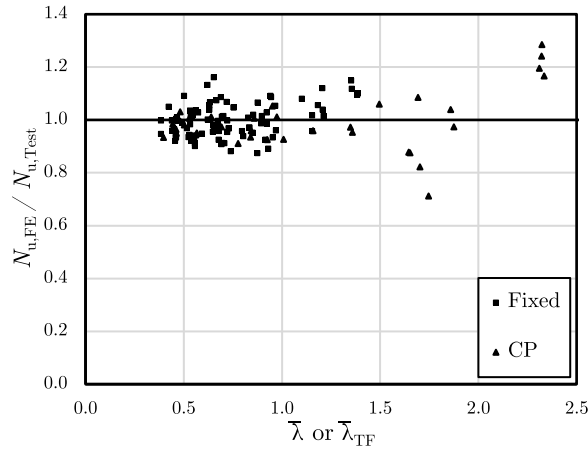


Figure 17: Ratio of FE model to test ultimate loads versus normalised slenderness $\bar{\lambda}$ or $\bar{\lambda}_{TF}$, whichever is higher (i.e. critical).

the test and FE ultimate loads for both fixed-ended (F) and cylindrically-pinned (CP) boundary conditions (B/C) and both considered manufacturing processes (hot-rolled and cold-formed), with a mean $N_{u,FE}/N_{u,Test}$ ratio (where $N_{u,FE}$ and $N_{u,Test}$ are the FE and test ultimate loads, respectively) of 1.00 and a coefficient of variation (CoV) of 0.08. There are a few test results that are less accurately predicted by the FE model - this is attributed to possible deviation between imperfections that were present in the physical tests and those assumed in the FE model.

Comparisons of typical load-deformation curves and deformed shapes, shown for Specimen 1 tested in the current study, are presented in Figs. 18 and 19, respectively, illustrating good agreement between test and FE results. Note that the FE model presented in Fig. 19b was created using the measured initial imperfections

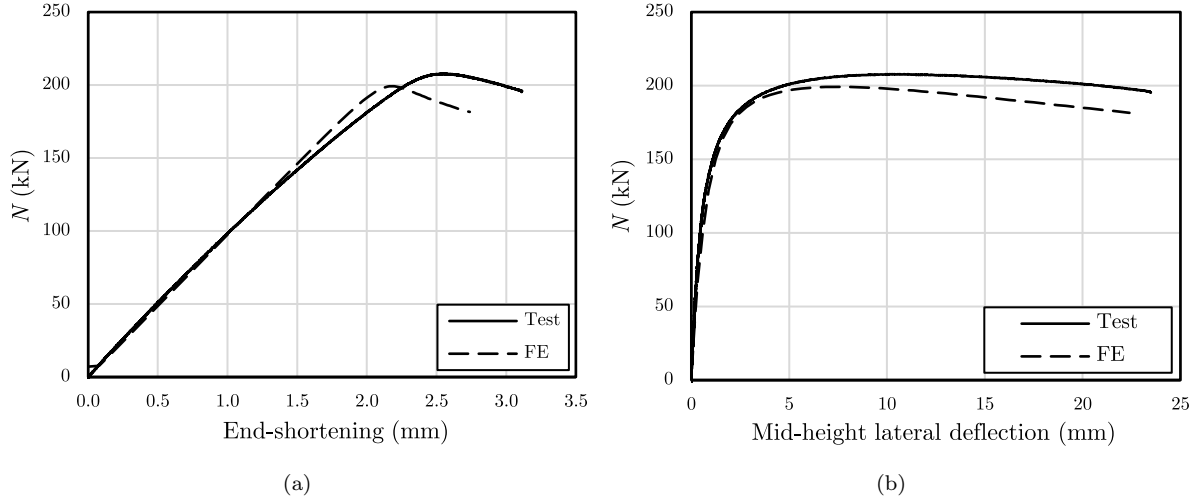


Figure 18: Comparisons of typical load N versus (a) end-shortening and (b) mid-height lateral deflection curves from a column buckling test and FE model for the specimen tested in the present study with $L = 2500$ mm.

(amplitude and direction); hence, the direction of buckling matched that observed in the tested specimen shown in Fig. 19a. The described comparisons demonstrate the ability of the developed FE models to predict the physical behaviour observed in the experiments on stainless steel angle section columns and the suitability of their use in a parametric study.

4.3. Parametric study

Following validation of the numerical models, a parametric study was conducted. The FE models were used to generate additional data for the three main families (austenitic, ferritic and duplex) of hot-rolled and cold-formed stainless steel angle section columns, considering a wide range of cross-section geometries and member lengths. The two-stage Ramberg–Osgood material model was used in combination with the standardised material properties proposed by [38]. To represent typical grades of austenitic, ferritic and duplex stainless steel, the values of $f_y = 280$ N/mm² and $f_u = 580$ N/mm², $f_y = 320$ N/mm² and $f_u = 480$ N/mm², and $f_y = 530$ N/mm² and $f_u = 770$ N/mm² [38], respectively, were employed; a Young’s modulus of 200000 N/mm² was used for all grades. For the cold-formed members, an enhanced strength, calculated using the predictive model presented in [39], was applied to the corner region of the angles. Leg widths b of 150 mm, 100 mm and 50 mm were analysed, while the thicknesses t and member lengths L were varied to evaluate a spectrum of normalised slenderness $\bar{\lambda}$ or $\bar{\lambda}_{TF}$ from 0.35 to 3.0, and $N_{cr,TF}/N_{cr,F,v}$ ratios from 0.01 to 4.5. The b/t ratios ranged between 5 and 50, while the L/b ratios ranged between 1 and 100. A summary of the ranges of the key varied parameters is presented in Table 7. A total of 1146 FE results was generated. The parametric study re-

Table 7: Summary of ranges of key varied parameters in FE parametric study.

Critical buckling mode	b (mm)	t (mm)	L (mm)	b/t	L/b	$N_{cr,TF}/N_{cr,F,v}$	λ_{TF} or λ
Torsional-flexural	50	2–10	50–1750	5–25	1–35	0.01–1.00	0.35–1.00
	100	3–5	400–7000	20–33	4–70	0.01–1.00	1.35–2.00
	150	3–4	1200–15000	38–50	8–100	0.01–1.00	2.35–3.00
Minor-axis flexural	50	4–8	700–5000	6–12	14–100	1.00–4.50	0.40–3.00
	100	3–6	4000–10000	16–33	40–100	1.00–3.00	1.00–2.65

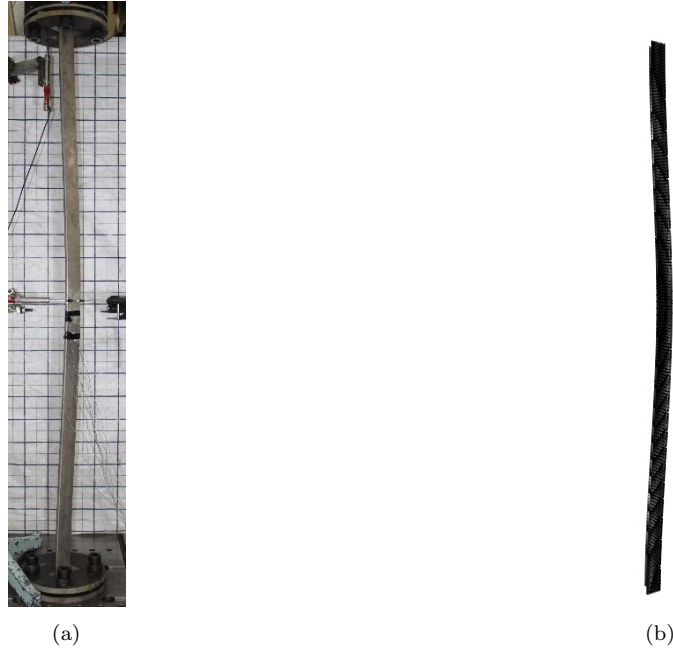


Figure 19: Comparison of typical column buckling deformed shape between (a) test and (b) FE model for specimen tested in the present study with $L = 2500$ mm.

sults for the fixed-ended hot-rolled and cold-formed stainless steel equal-leg angle section columns are presented in Figs. 20 and 21, respectively, where the buckling reduction factor $\chi = N_u/Af_y$ (where N_u is the ultimate

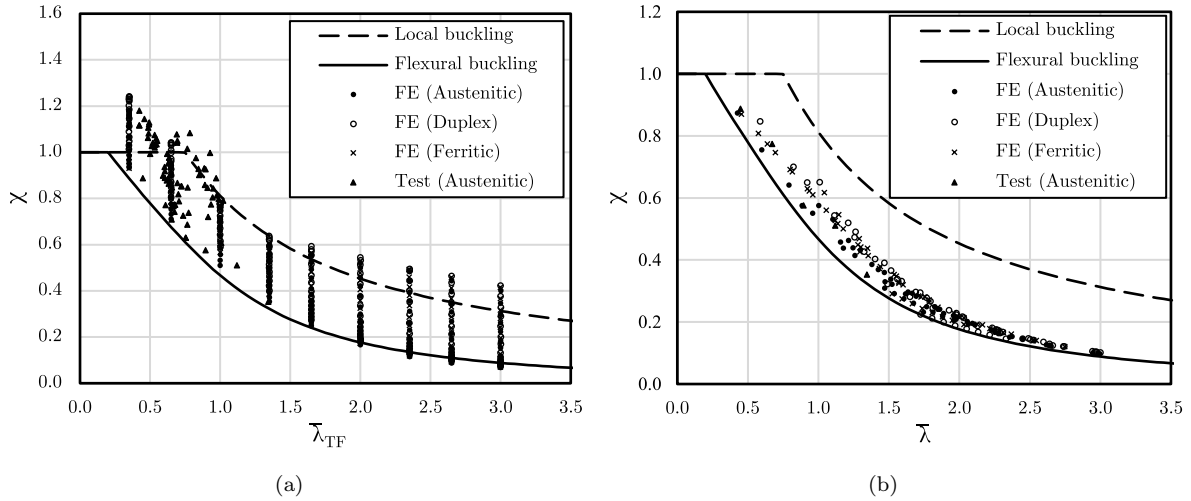


Figure 20: Comparison of test and FE results for fixed-ended hot-rolled austenitic, ferritic and duplex stainless steel equal-leg angle section columns where (a) torsional-flexural buckling and (b) minor-axis flexural buckling are critical against EC3 buckling curves.

load obtained from the FE models) is plotted against the corresponding normalised slenderness. Note that the weighted average yield stress [40] is used in the normalisation for the cold-formed angles. The results for the cases where torsional-flexural buckling is critical (i.e. $N_{cr,TF} \leq N_{cr,F,v}$) are shown in Figs. 20a and 21a, with the normalised slenderness being $\bar{\lambda}_{TF}$, while the results for the cases where minor-axis flexural buckling is critical (i.e. $N_{cr,F,v} < N_{cr,TF}$) are shown in Figs. 20b and 21b, with the normalised slenderness being $\bar{\lambda}$. As explained in Section 2, for equal-leg angle section columns, since the torsional and local buckling modes are essentially the same, use of both a local and member buckling reduction factor amounts to double-counting; therefore, in Figs.

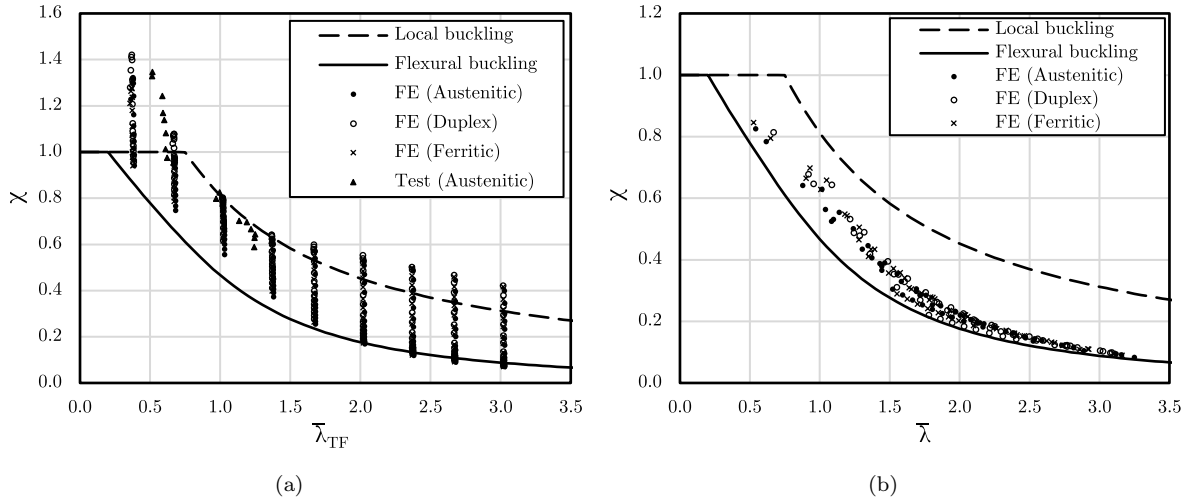


Figure 21: Comparison of FE results for fixed-ended cold-formed austenitic, ferritic and duplex stainless steel equal-leg angle section columns where (a) torsional-flexural buckling and (b) minor-axis flexural buckling are critical against EC3 buckling curves.

20a and 21a, the torsional-flexural member slenderness, which approaches the torsional (or local) slenderness for short members where $N_{cr} \approx N_{cr,T}$, is used on the horizontal axis and the gross area is employed in the normalisation of the strength of the columns. For reference, the Eurocode 3 (EC3) local (for single outstand flanges from EN 1993-1-4 [5]) and flexural (from EN 1993-1-4 [5]) buckling curves are also shown, corresponding to the torsional and minor-axis flexural buckling modes, respectively, for angle section columns. The results are analysed and discussed in the following section.

5. Analysis and discussion of results

The assembled experimental and numerical data are analysed and discussed in this section. It can be seen from Figs. 20a and 21a that, where torsional-flexural buckling is critical, stainless steel angle section columns with the same slenderness can have very different buckling reduction factors. This contrasts with the existing EC3 design approach, where angle section columns with the same member slenderness have the same member buckling reduction factor because, regardless of the critical buckling mode (torsional-flexural or minor-axis flexural), the flexural buckling curve is always applied. Similar observations have been made by Behzadi-Sofiani et al. [2] and [3] for steel equal-leg angle section columns. The vertical spread of data in Figs. 20a and 21a is caused by the changing post-buckling stability with the $N_{cr,TF}/N_{cr,F,v}$ ratio. The numerically-derived [2] ratios of the immediate elastic post-buckling axial stiffness to the initial axial stiffness k_{pb}/k_{in} for a series of fixed-ended angle section columns are plotted against $N_{cr,TF}/N_{cr,F,v}$ in Fig. 22. The k_{pb}/k_{in} ratios can be seen to reduce from 0.44 (which is equal to the theoretical k_{pb}/k_{in} ratio for plates with one edge simply-supported and one edge free, failing by local buckling, or equal-leg angles, failing by purely torsional buckling [41]) to zero (which corresponds to the neutral post-buckling stability associated with flexural buckling) as $N_{cr,TF}/N_{cr,F,v}$ varies between zero and 1.0.

When torsional-flexural buckling is critical (i.e. $N_{cr,TF} \leq N_{cr,F,v}$) – see Figs. 20a and 21a, for the same slenderness $\bar{\lambda}_{TF}$, the $N_{cr,TF}/N_{cr,F,v}$ ratio can vary between zero (indicating torsional-dominant buckling with

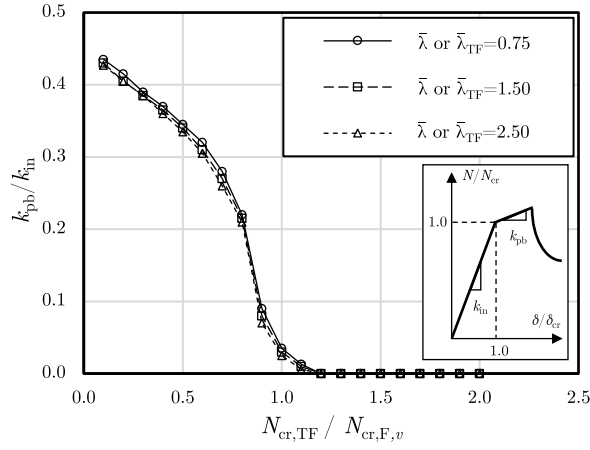


Figure 22: Relationship between ratio of immediate elastic post-buckling axial stiffness k_{pb} to initial axial stiffness k_{in} and $N_{cr,TF}/N_{cr,F,v}$ for equal-leg angle section columns with three slenderness values: $\bar{\lambda}$ or $\bar{\lambda}_{TF} = 0.75, 1.5$ and 2.5 [2], highlighting transition in post-buckling stability from stable to neutral as $N_{cr,TF}/N_{cr,F,v}$ increases. In the inset diagram, δ is the axial displacement and δ_{cr} is the axial displacement at N_{cr} .

plate-like behaviour and corresponding to the high data points on the graph) and 1.0 (denoting the transition point from torsional-flexural buckling to minor-axis flexural buckling and corresponding to the low data points on the graph). For $N_{cr,TF}/N_{cr,F,v}$ ratios greater than 1.0, minor-axis flexural buckling becomes critical, for which the results are presented in Figs. 20b and 21b.

In Figs. 20a and 21a, the top data points correspond to the stainless steel angle section columns with the lowest $N_{cr,TF}/N_{cr,F,v}$ ratio (0.01) and lie in the region of the local buckling curve, set out in EN 1993-1-4 [5], which reflects the beneficial stable post-buckling behaviour of plates; as this ratio increases, the reduction factor decreases and converges to the flexural buckling curve, which reflects the approximately neutral post-buckling stability of columns failing in this mode. For low $N_{cr,TF}/N_{cr,F,v}$ ratios, the reduction factor reduces at a higher rate; this shows that even a small participation of the flexural buckling mode can reduce the ultimate strength significantly. For $N_{cr,TF}/N_{cr,F,v}$ ratios greater than 1.0, the corresponding reduction factors generally follow the column flexural buckling curve, as shown in Figs. 20b and 21b.

6. Design of stainless steel angle section columns to Eurocode 3

Stainless steel members are designed according to the supplementary rules set out in EN 1993-1-4 [5]. Stainless steel angle sections are deemed to be Class 1–3 (i.e. not susceptible to local buckling) if the following criteria are satisfied:

$$\frac{b}{t} \leq 15\epsilon \quad (13)$$

$$\frac{h+b}{2t} \leq 11.5\epsilon \quad (14)$$

where h is the width of the longer leg in unequal-leg angles. Beyond these limits, cross-sections are Class 4 and the effective area A_{eff} formulae given in EN 1993-1-4 [5] apply. However, EN 1993-1-4 does not make a clear distinction between equal- and unequal-leg angles. If both criteria (i.e. Eqs. (13) and (14)) are applied to equal-leg angles, then Eq. (14) is always critical. As a result, equal-leg angle sections in which the b/t ratio

exceeds 11.5ϵ are deemed to be Class 4, though the local buckling reduction factors ρ given in EN 1993-1-4 do not fall below unity until higher b/t ratios are reached – this is a clear inconsistency. In EN 1993-3-1 [42], the ratio $(b - 2t)/t$ is used instead of b/t for equal-leg angles.

The calculation of normalised member slenderness is based on the elastic buckling load for torsional (T), flexural (F) and torsional-flexural (TF) buckling in EN 1993-1-4. The design member buckling resistance $N_{b,Rd}$ is given by Eq. (15):

$$N_{b,Rd} = \frac{\chi A f_y}{\gamma_{M1}} \quad (15)$$

where A is the cross-sectional area, which is replaced by the effective area A_{eff} for Class 4 cross-sections, χ is the buckling reduction factor from the relevant member buckling curve and γ_{M1} is the partial factor for the resistance of members to instability, equal to 1.1 for stainless steel members. The buckling reduction factor χ is given by Eq. (16):

$$\chi = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}_i^2}} \quad \text{but} \quad \chi \leq 1.0 \quad (16)$$

where $\bar{\lambda}_i$ is the relevant normalised slenderness, defined as:

$$\bar{\lambda}_i = \sqrt{\frac{A f_y}{N_{cr,i}}} \quad i=\{T, F, TF\} \quad (17)$$

in which $N_{cr,i}$ is the elastic buckling load for the relevant mode i and ϕ is an auxiliary parameter used to determine the buckling reduction factor χ , given by:

$$\phi = 0.5 [1 + \eta + \bar{\lambda}_i^2] \quad (18)$$

where η is the generalised initial imperfection factor, defined as:

$$\eta = \alpha (\bar{\lambda}_i - \bar{\lambda}_0) \quad (19)$$

in which α is the imperfection factor and $\bar{\lambda}_0$ is the limiting slenderness. In prEN 1993-1-4, $\bar{\lambda}_0 = 0.2$ and $\alpha = 0.76$ is specified for both hot-rolled and cold-formed stainless steel angle section columns. Comparisons of the test and FE ultimate capacities N_u with the resistances predicted by EC3 $N_{b,EC3}$ are made in Table 8 and Figs. 23 and 24. In Table 8, the results are presented by manufacturing process, stainless steel family and critical buckling mode. In the figures, a distinction is made between the hot-rolled (Fig. 23) and cold-formed (Fig. 24) stainless steel angles, and between the columns where torsional-flexural (Figs. 23a and 24a) and minor-axis flexural (Figs. 23b and 24b) buckling are critical. From the comparisons, it is clear that the current EC3 provisions are extremely conservative in predicting the strength of stainless steel equal-leg angle section columns where torsional-flexural buckling is critical. This can be explained, as described by Dinis et al. [43] and Behzadi-Sofiani et al. [2], by the double-counting of the same buckling mode in the local and member buckling checks and the failure to recognise the stable post-buckling behaviour in the member-level buckling

Table 8: Summary of comparisons between test/FE and EC3 ultimate capacities for fixed-ended stainless steel equal-leg angle section columns.

Source	Manufacturing process	Stainless steel family	Evaluation parameter	$N_u/N_{b,EC3}$		
				Torsional-flexural	Flexural	Total
FE	Hot-rolling	Austenitic	Mean	1.85	1.16	1.65
			CoV	0.50	0.05	0.51
		Duplex	Mean	2.04	1.21	1.80
			CoV	0.49	0.08	0.51
	Cold-forming	Ferritic	Mean	1.94	1.22	1.73
			CoV	0.50	0.06	0.51
		Austenitic	Mean	1.94	1.23	1.73
			CoV	0.48	0.06	0.49
		Duplex	Mean	2.10	1.24	1.85
			CoV	0.48	0.09	0.51
Test	Hot-rolling	Austenitic	Mean	1.49	1.11	1.46
			CoV	0.19	0.07	0.20
	Cold-forming	Austenitic	Mean	1.83	-	1.83
			CoV	0.14	-	0.14

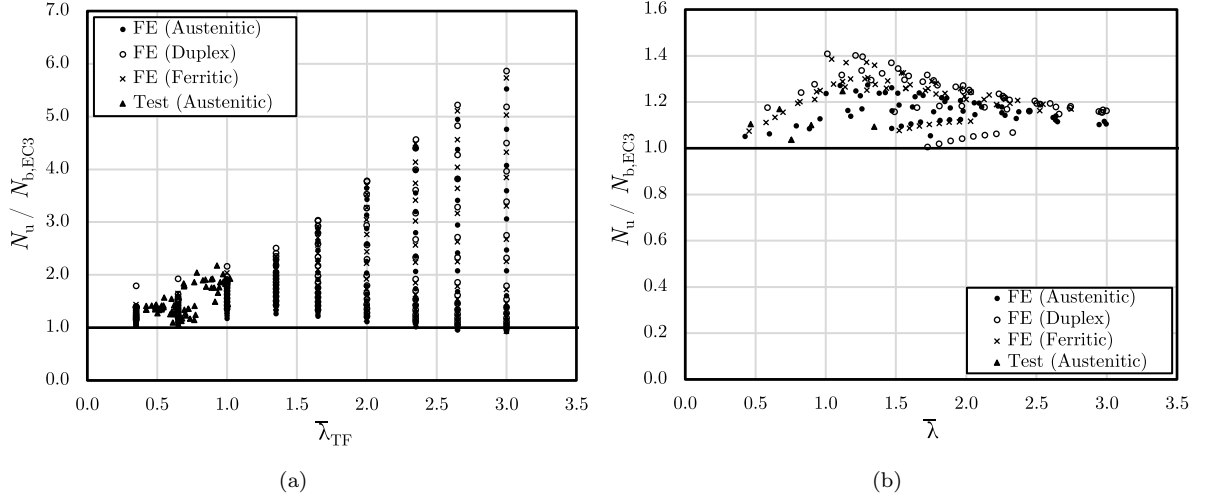


Figure 23: Comparisons between test/FE and EC3 ultimate capacities for fixed-ended hot-rolled austenitic, duplex and ferritic stainless steel equal-leg angle section columns where (a) torsional-flexural and (b) minor-axis flexural buckling are critical.

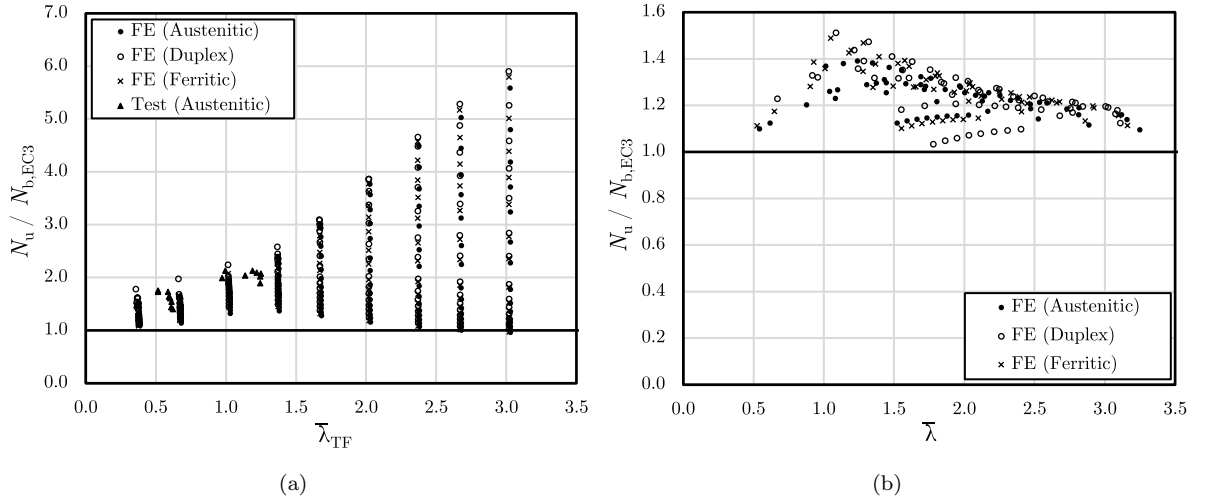


Figure 24: Comparisons between test/FE and EC3 ultimate capacities for fixed-ended cold-formed austenitic, duplex and ferritic stainless steel equal-leg angle section columns where (a) torsional-flexural and (b) minor-axis flexural buckling is critical.

curves for torsional-flexural buckling. New design proposals are presented in the subsequent section to address and resolve this issue.

7. New design proposals

Mirroring recent proposals [2] for the design of fixed-ended steel equal-leg angle section columns, new proposals for the design of fixed-ended stainless steel equal-leg angle section columns are presented and assessed in the current section.

7.1. Torsional-flexural buckling ($N_{cr,TF} \leq N_{cr,F,v}$)

The proposed design buckling resistance $N_{b,Rd}$ of stainless steel equal-leg angle section columns where $N_{cr,TF}/N_{cr,F,v} \leq 1.0$ is given thus:

$$N_{b,Rd} = \frac{\chi_{TF} A f_y}{\gamma_{M1}} \quad (20)$$

noting that the gross area is used for all classes of cross-section to eliminate the effective double-counting of the torsional-dominant buckling mode in the cross-section (i.e. local) and member level stability checks, as occurs in the current EC3 design provisions. In Eq. (20):

$$\chi_{TF} = \chi_F + \Delta_F (\chi_T - \chi_F) \quad (21)$$

in which the torsional buckling reduction factor χ_T is given by:

$$\chi_T = \frac{\bar{\lambda}_{TF} - 0.188}{\bar{\lambda}_{TF}^2} \quad \text{but} \quad \chi_T \leq 1.0 \quad (22)$$

the flexural buckling reduction factor χ_F is given by:

$$\chi_F = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}_{TF}^2}} \quad \text{but} \quad \chi_F \leq 1.0 \quad (23)$$

and Δ_F is given thus:

$$\Delta_F = \left(1 - \frac{N_{cr,TF}}{N_{cr,F,v}} \right)^p \quad (24)$$

where

$$p = \begin{cases} 2.0 \bar{\lambda}_{TF} & \text{for } \bar{\lambda}_{TF} \leq 2.0 \\ 2.93 \bar{\lambda}_{TF}^{0.45} & \text{for } \bar{\lambda}_{TF} > 2.0 \end{cases} \quad (25)$$

with the torsional-flexural slenderness $\bar{\lambda}_{TF}$ and ϕ being given thus:

$$\bar{\lambda}_{TF} = \sqrt{\frac{A f_y}{N_{cr,TF}}} \quad (26)$$

$$\phi = 0.5 \left[1 + \alpha \beta (\bar{\lambda}_{TF} - \bar{\lambda}_0)^\beta + \bar{\lambda}_{TF}^2 \right] \quad (27)$$

For both hot-rolled and cold-formed stainless steel angles, the proposed values for β and the limiting slenderness $\bar{\lambda}_0$ are 1.45 and 0.2, respectively. For the imperfection factor α , values of 0.6 and 0.49 are recommended for hot-rolled and cold-formed stainless steel angles, respectively. The value of 0.6 is also used for laser-welded sections in prEN 1993-1-4 [24, 26], but could be relaxed to 0.76 if retention of the traditional buckling curves (i.e. curve d in this case) was desirable. Note though, for $\bar{\lambda}_{TF}$ and $\bar{\lambda}$ values greater than about 1.0, that despite the lower value for the imperfection factor α , with the newly introduced parameter β , the proposed buckling curves lie below the existing EN 1993-1-4 buckling curve in the region $\frac{N_{cr,TF}}{N_{cr,F,v}} = 1.0$, reflecting the influence of mode interaction between torsional-flexural and minor-axis flexural buckling. The variation of χ with $\bar{\lambda}_{TF}$ and $\frac{N_{cr,TF}}{N_{cr,F,v}}$ is shown in Fig. 25 for three sample values of $\frac{N_{cr,TF}}{N_{cr,F,v}} = 0.1$, $\frac{N_{cr,TF}}{N_{cr,F,v}} = 0.2$ and $\frac{N_{cr,TF}}{N_{cr,F,v}} = 1.0$ together with

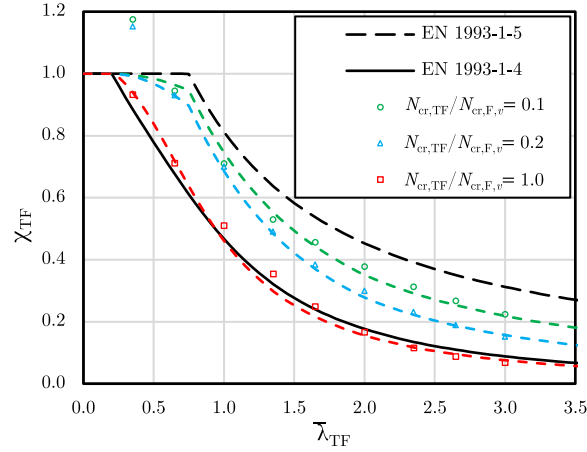


Figure 25: Torsional-flexural reduction factor χ_{TF} versus torsional-flexural slenderness $\bar{\lambda}_{TF}$ for fixed-ended hot-rolled stainless steel equal-leg angle section columns.

the FE results from the parametric study for columns with $\frac{N_{cr,TF}}{N_{cr,F,v}}$ ratios of similar values.

7.2. Minor-axis flexural buckling ($N_{cr,F,v} < N_{cr,TF}$)

The design buckling resistance $N_{b,Rd}$ for equal-leg angle section columns where $N_{cr,TF}/N_{cr,F,v} > 1.0$ is obtained thus:

$$N_{b,Rd} = \frac{\chi_F A f_y}{\gamma_{M1}} \quad (28)$$

where

$$\chi_F = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}^2}} \quad \text{but} \quad \chi_F \leq 1.0 \quad (29)$$

in which the normalised slenderness $\bar{\lambda}$, ϕ and the generalised initial imperfection factor η are given by:

$$\bar{\lambda} = \sqrt{\frac{A f_y}{N_{cr,F,v}}} \quad (30)$$

$$\phi = 0.5 [1 + \eta + \bar{\lambda}^2] \quad (31)$$

$$\eta = \alpha \beta (\bar{\lambda} - \bar{\lambda}_0)^\beta \quad (32)$$

with β being a factor allowing for the influence of interactive buckling:

$$\beta = 1.9 - 0.45 \frac{N_{cr,TF}}{N_{cr,F,v}} \quad \text{but} \quad 1.0 \leq \beta \leq 1.45 \quad \text{for cold-formed angles.} \quad (33)$$

As $N_{cr,TF}/N_{cr,F,v}$ increases from 1.0 to 2.0, β converges to 1.0 as the influence of interactive buckling diminishes. For $N_{cr,TF}/N_{cr,F,v} > 2.0$, the original flexural buckling curve applies (i.e. $\beta = 1.0$). At $N_{cr,TF}/N_{cr,F,v} = 1.0$ (i.e. at the transition between torsional-flexural and minor-axis flexural buckling), β reaches its upper bound value of 1.45. This value is retained for all cases in which torsional-flexural buckling is critical (i.e. $N_{cr,TF}/N_{cr,F,v} \leq 1.0$). Note that, unlike in the case of torsional-flexural buckling, the gross area A is replaced by the effective area A_{eff} [44] in Eqs. (28) and (30) for Class 4 cross-sections when minor-axis flexural buckling is critical since the issue of double-counting the torsional-dominant buckling mode no longer exists; A_{eff} is determined from EN 1993-1-4 [5]. However, for consistency with outstand elements in EN 1993-1-4, it is proposed that the current Class 3 slenderness limit of $b/t = 11.5\epsilon$ for equal-leg angle sections is changed to $b/t = 14\epsilon$. The imperfection factor and limiting slenderness remain as specified above: $\alpha = 0.6$ and $\alpha = 0.49$ for hot-rolled and cold-formed stainless steel angles, respectively, and $\bar{\lambda}_0 = 0.2$ for both hot-rolled and cold-formed stainless steel angles.

7.3. Assessment of design proposals

A summary of the comparisons of the test and FE capacities against the resistance predictions according to the new proposals $N_{b,prop}$ is presented in Table 9 and Figs. 26 and 27. By comparing Figs. 26a and 27a, and

Table 9: Summary of comparisons between test/FE capacities and resistance predictions obtained from proposed design method for fixed-ended stainless steel equal-leg angle section columns.

Source	Manufacturing process	Stainless steel family	Evaluation parameter	$N_u/N_{b,prop}$		
				Torsional-flexural	Flexural	Total
FE	Hot-rolling	Austenitic	Mean	1.09	1.13	1.10
			CoV	0.10	0.07	0.09
		Duplex	Mean	1.21	1.19	1.20
			CoV	0.10	0.08	0.10
		Ferritic	Mean	1.15	1.19	1.16
			CoV	0.11	0.07	0.10
	Cold-forming	Austenitic	Mean	1.12	1.14	1.13
			CoV	0.10	0.07	0.09
		Duplex	Mean	1.21	1.15	1.19
			CoV	0.10	0.08	0.10
		Ferritic	Mean	1.16	1.16	1.16
			CoV	0.10	0.08	0.10
Test	Hot-rolling	Austenitic	Mean	1.09	1.04	1.09
			CoV	0.10	0.06	0.10
	Cold-forming	Austenitic	Mean	1.14	-	1.14
			CoV	0.11	-	0.11

23a and 24a, it can be seen that the resistance predictions are significantly improved for stainless steel angles exhibiting torsional-flexural buckling using the new proposals relative to the current Eurocode 3 provisions. Comparing Figs. 26b and 27b, and 23b and 24b, the proposed resistance predictions can also be seen to be more accurate than those given by the current Eurocode 3 provisions for stainless steel angles failing by minor-axis

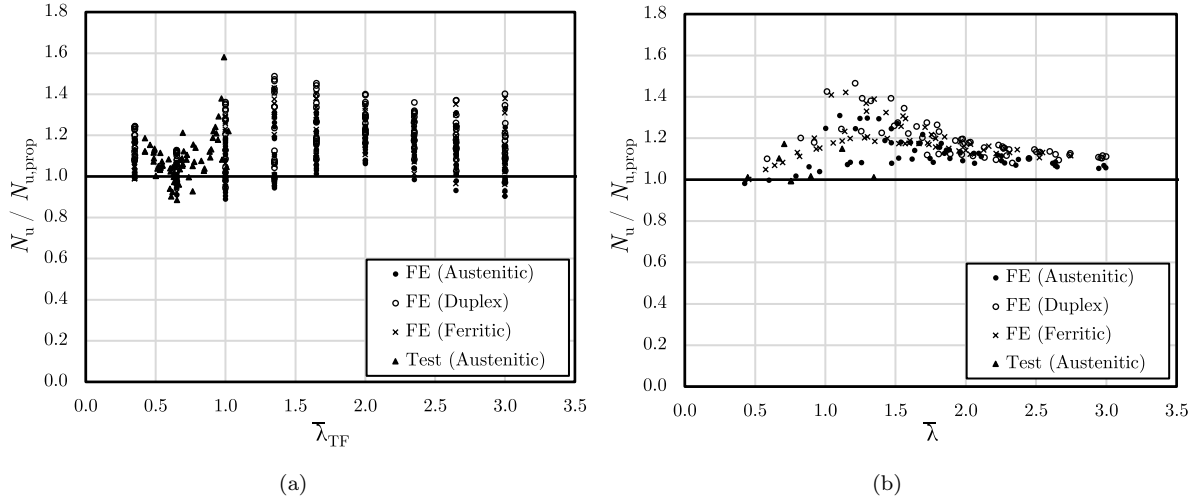


Figure 26: Comparisons of test and FE results against the new proposals for the design of fixed-ended hot-rolled austenitic, duplex and ferritic stainless steel equal-leg angle section columns for the cases of: (a) torsional-flexural buckling and (b) minor-axis flexural buckling being critical.

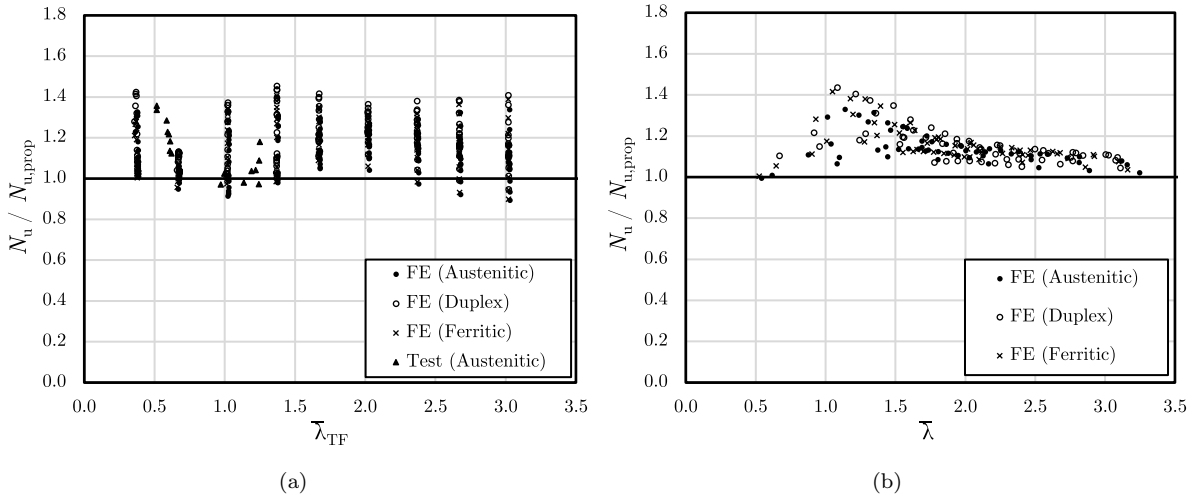


Figure 27: Comparisons of test and FE results against the new proposals for the design of fixed-ended cold-formed austenitic, duplex and ferritic stainless steel equal-leg angle section columns for the cases of: (a) torsional-flexural buckling and (b) minor-axis flexural buckling being critical.

flexural buckling. Note that the proposed resistance functions for torsional-flexural and minor-axis flexural buckling achieve continuity at the transition point between the two modes (i.e. when $N_{cr,TF}/N_{cr,F,v} = 1.0$).

7.4. Reliability analysis

Statistical analyses were performed in accordance with [45] to evaluate the reliability of the current EC3 [5] and proposed design approaches for fixed-ended stainless steel equal-leg angle section columns. As provided in Afshan et al. [46], the material over-strength factor $f_{y,m}/f_{y,n}$ (i.e. the ratio of the mean to the nominal yield strength) and the corresponding CoV of yield strength V_{f_y} were taken as 1.3 and 0.06 for austenitic, 1.1 and 0.03 for duplex, and 1.2 and 0.045 for ferritic stainless steel, while the CoV of the Young's modulus was taken as 0.03. The CoV of the cross-sectional area V_A was calculated based on the parameters of the geometric dimensions provided in Annex E of [47] and following the procedure outlined in [46], resulting in a value of V_A equal to 0.03. Following Annex E of [47], the mean values of the Young's modulus and all the geometric

properties were taken equal to the nominal values. A summary of the reliability analysis procedure is provided in [2]. The key reliability analysis results are presented in Tables 10 and 11 for the current EC3 and the new

Table 10: Reliability analysis results for EC3.

Source	Manufacturing process	Grade	Buckling mode	$k_{d,n}$	b	V_δ	V_r	γ_{M1}^*	$\frac{\gamma_{M1}^*}{\gamma_{M1}}$	f_a	$\frac{\gamma_{M1}^*}{\gamma_{M1}} \frac{f_a}{f_a}$
FE	Hot-rolling	Austenitic	Torsional-flexural	3.142	1.846	0.425	0.428	3.29	2.99	1.27	2.36
			Minor-axis flexural	3.142	1.164	0.049	0.072	1.02	0.92	1.16	0.80
		Duplex	Torsional-flexural	3.142	2.044	0.425	0.426	3.22	2.93	1.27	2.31
			Minor-axis flexural	3.142	1.212	0.082	0.098	1.10	1.00	1.18	0.85
		Ferritic	Torsional-flexural	3.142	1.939	0.433	0.435	3.33	3.03	1.27	2.39
			Minor-axis flexural	3.142	1.225	0.065	0.084	1.01	0.92	1.17	0.79
	Cold-forming	Austenitic	Torsional-flexural	3.142	1.942	0.413	0.416	3.02	2.75	1.27	2.17
			Minor-axis flexural	3.142	1.231	0.063	0.083	1.00	0.91	1.17	0.78
		Duplex	Torsional-flexural	3.142	2.102	0.422	0.424	3.10	2.82	1.27	2.23
			Minor-axis flexural	3.142	1.238	0.086	0.102	1.10	1.00	1.18	0.84
		Ferritic	Torsional-flexural	3.142	2.003	0.423	0.425	3.13	2.84	1.27	2.25
			Minor-axis flexural	3.142	1.257	0.079	0.095	1.04	0.94	1.18	0.80
Test	Hot-rolling	Austenitic	Torsional-flexural	3.231	1.494	0.180	0.188	1.36	1.24	1.25	0.99
			Minor-axis flexural	3.231	1.113	0.142	0.051	1.20	1.10	1.14	0.96
	Cold-forming	Austenitic	Minor-axis flexural	3.848	1.826	0.142	0.152	1.31	1.19	1.23	0.97

Table 11: Reliability analysis results for the new design proposals.

Source	Manufacturing process	Grade	Buckling mode	$k_{d,n}$	b	V_δ	V_r	γ_{M1}^*	$\frac{\gamma_{M1}^*}{\gamma_{M1}}$	f_a	$\frac{\gamma_{M1}^*}{\gamma_{M1}} \frac{f_a}{f_a}$
FE	Hot-rolling	Austenitic	Torsional-flexural	3.142	1.094	0.099	0.119	1.15	1.05	1.20	0.87
			Minor-axis flexural	3.142	1.130	0.067	0.086	1.10	1.00	1.17	0.85
		Duplex	Torsional-flexural	3.142	1.206	0.102	0.109	1.12	1.02	1.19	0.86
			Minor-axis flexural	3.142	1.188	0.074	0.092	1.10	1.00	1.18	0.85
		Ferritic	Torsional-flexural	3.142	1.147	0.108	0.120	1.16	1.06	1.20	0.88
			Minor-axis flexural	3.142	1.187	0.069	0.088	1.06	0.96	1.17	0.82
	Cold-forming	Austenitic	Torsional-flexural	3.142	1.121	0.098	0.118	1.12	1.02	1.20	0.85
			Minor-axis flexural	3.142	1.136	0.064	0.085	1.10	1.00	1.17	0.85
		Duplex	Torsional-flexural	3.142	1.209	0.100	0.107	1.11	1.01	1.19	0.85
			Minor-axis flexural	3.142	1.153	0.073	0.092	1.14	1.04	1.18	0.88
		Ferritic	Torsional-flexural	3.142	1.155	0.103	0.116	1.14	1.04	1.20	0.87
			Minor-axis flexural	3.142	1.161	0.073	0.091	1.11	1.01	1.18	0.86
Test	Hot-rolling	Austenitic	Torsional-flexural	3.231	1.092	0.096	0.115	1.07	0.97	1.19	0.81
	Cold-forming		Minor-axis flexural	3.231	1.040	0.057	0.076	1.05	0.96	1.16	0.82
		Austenitic	Minor-axis flexural	3.848	1.135	0.112	0.129	1.16	1.05	1.21	0.87

design proposals, respectively, where the design fractile factor $k_{d,n}$, the CoV of the test and FE resistances relative to the predictions from the resistance model V_δ , and the combined CoV, incorporating the variability of the resistance model and the basic variables V_r , were calculated in accordance with [45], γ_{M1}^* is the required value of the partial safety factor taken as the mean value for all test and FE results and determined following the approach in [46], γ_{M1} is the target partial safety factor value (equal to 1.1) and f_a is the acceptance limit, given thus [48]:

$$f_a = 1.133 + 0.825(V_r - 0.04) \quad \text{but} \quad 1.133 \leq f_a \leq 1.265 \quad (34)$$

and the ratio $\gamma_{M1}^*/\gamma_{M1}$ expressed thus:

$$\gamma_{M1}^*/\gamma_{M1} \leq f_a. \quad (35)$$

A graphical comparison of $\gamma_{M1}^*/\gamma_{M1}$ and the recommended acceptance limit f_a is presented in Fig. 28 for the new design method. The current EC3 design expressions clearly lead to unacceptably high γ_{M1}^* values, owing principally to the excessively high scatter of the resistance predictions. The proposed design method,

on the other hand, leads to $(\gamma_{M1}^*/\gamma_{M1})/f_a$ ratios less than 1.0 for both hot-rolled and cold-formed members and are therefore considered to satisfy the reliability requirements (i.e. the $(\gamma_{M1}^*/\gamma_{M1})$ values are lower than the acceptance limit f_a). Note that, even with the proposed design rules, owing to the relatively high scatter associated with the sensitive buckling response of angle section columns compared to, for example, I-section and hollow section [46] columns failing by flexural buckling, the mean resistance predictions need to be somewhat higher (i.e. higher b values) than is customary.

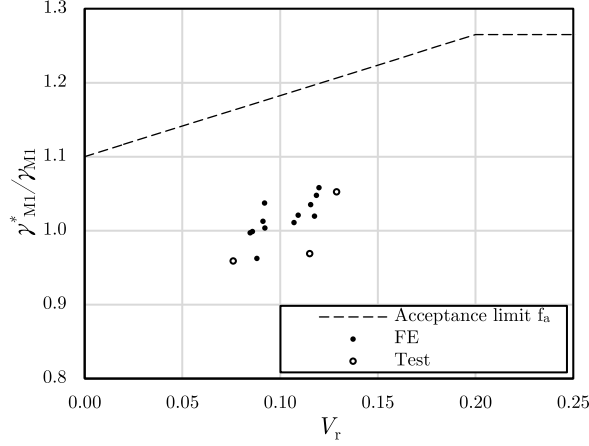


Figure 28: Comparison of the $\gamma_{M1}^*/\gamma_{M1}$ ratios with the acceptance limit f_a for the new design proposals.

8. Conclusions

A comprehensive study into the behaviour and stability design of fixed-ended stainless steel equal-leg angle section members subjected to axial compression has been presented. A series of experiments on stainless steel angle section columns, together with supporting material tests and initial geometric imperfection measurements, was first reported. Finite element models were then developed and validated against the reported test results, as well as existing experimental data from the literature. Good agreement was shown between the test and numerical results. A parametric study was subsequently conducted on both hot-rolled and cold-formed angle section columns in austenitic, duplex and ferritic stainless steel with fixed-ended support conditions covering a wide spectrum of cross-section and member geometries. A total of 1146 numerical results was generated. The numerical data were combined with existing test results from the literature and used to assess the current design provisions in EN 1993-1-4 [5] for stainless steel equal-leg angle section columns. The current Eurocode 3 design provisions are shown to be conservative in cases where the torsional-flexural buckling mode is critical due to the effective double-counting of the same buckling mode in the cross-section and member checks alongside the failure to recognise the stable post-buckling characteristic of the torsional-flexural mode in the member buckling curves. New design provisions are developed for fixed-ended stainless steel equal-leg angle section columns, covering both torsional-flexural and minor-axis flexural buckling that capture the influence of the transition in post-buckling behaviour from stable to neutral with increasing $\frac{N_{cr,TF}}{N_{cr,F,v}}$ ratios and the effects of interactive buckling between the torsional-flexural and minor-axis flexural buckling modes in the region where $\frac{N_{cr,TF}}{N_{cr,F,v}} \approx 1.0$. The new design rules provide dramatic improvements in both the accuracy and consistency of

413 strength predictions for all grades of stainless steel equal-leg angle section columns with fixed-ended support
414 conditions, and satisfy the Eurocode reliability requirements.

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